The Model in Three Pages

Multinational ownership, market power, and the measurement of foreign profits

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1 The question, and why it needs this model

A large share of developing-country exports is shipped by subsidiaries of international groups headquartered elsewhere — often in the country imposing the tariff. **How much of the profit earned in those markets belongs to that country?** The obvious calculation is total profit times the share of firms owned. Proposition 1 shows this is wrong, and gives the exact error.

Three features are needed at once, and no existing framework has all three: markups that *rise with firm size* (otherwise ownership is a pure transfer — Proposition 2); *multinational production* (headquarters in one country, factory in a second, sales to a third); and *entry* (which factories exist must respond to policy). Combining them creates a multiplicity problem, which Theorem 2 solves.

2 Where the model comes from

The market structure comes from Atkeson and Burstein (2008): two layers of competition, where goods compete closely inside a sector and loosely across sectors, which is what makes a firm's markup depend on how big it is. The one change is that the markup is computed at the level of the *owner* rather than the factory. A group that owns several plants in the same market internalises the competition among them, so what sets its markup is its combined share, not each plant's — the mechanism Yang (2023) calls cannibalisation, and the reason ownership shows up in prices at all. With constant markups the entire result vanishes, which is why monopolistic competition is not an option here. Which factories exist is then an outcome rather than a list, and the granular machinery for that is Gaubert and Itskhoki (2021): capability drawn from a Pareto distribution, and a pool of potential parents that grows with country size, so the concentration of parent firms in a handful of rich countries comes from a bigger pool having a better maximum rather than from assuming that rich countries breed better firms. Entry is settled by a theorem rather than by an assumed order of moves, which matters because an order of moves is not a property of the model.

The multinational side comes from Ramondo and Rodríguez-Clare (2013): producing away from head office is less efficient, a friction measured for the car industry by Head and Mayer (2019). Tintelnot (2017) supplies the geography — shipping is counted from where a good is *made* rather than from head office, so one factory can serve many countries, and serving each of them costs a fixed amount, which is what turns location into a decision. Caliendo and Parro (2015) supply the input–output loop, so that firms buy from each other at the prices prevailing where they sit. Head-office intensity is Antràs (2003) and Antràs and Helpman (2004), but used on the *capability* side rather than the cost side: a group's head office is what makes its factories good, not a service each factory buys at its parent's wage. Complexity therefore raises the penalty on a stand-alone local firm and raises the return to a group's capability, but it never puts a foreign wage into a domestic cost — a choice that turns out to decide whether the model has one solution, and which improves the fit besides: it raises the Fact 2 gradient from +0.33 to +0.42 against a target of +0.43 and makes the fitted multinational productivity edge unnecessary, so four internally calibrated parameters become three. The version where head-office services are bought at the parent's wage is supported and reported alongside. What is not borrowed from anyone: ownership is at the firm level, θ, so who owns which factory is data rather than a country-level average — and that is the whole point.

3 One market

A *market* is one product in one country. *Factories* i sell into it; each belongs to a *group* g, possibly with factories in several countries. S_g is the group's share of market sales; σ and η are how easily buyers substitute within and away from the market, $\sigma > \eta \geq 1$.

The mechanism: a group competes with itself. When a group expands one factory it takes sales from rivals *and from its own other factories*. It counts the second loss, because it collects both profits. So it holds back — and the larger its share, the more it holds back:

$$\text{quantity competition: } \frac{1}{\mu(S)} = 1 - \frac{1-S}{\sigma} - \frac{S}{\eta}; \qquad \text{price competition: } \mu(S) = \frac{\varepsilon(S)}{\varepsilon(S)-1}, \quad \varepsilon(S) = \sigma - (\sigma-\eta)S. \quad (1)$$

The markup μ is a weighted average of two elasticities: the group faces the tough elasticity σ on the $1-S$ it does not own, and the gentle η on the S it does, where expanding only cannibalises itself. Both forms run from $\sigma/(\sigma-1)$ at $S=0$ to $\eta/(\eta-1)$ at $S=1$. *All results below hold for any μ satisfying:*

Assumption 1. μ is continuously differentiable and strictly increasing on $[0,1)$, with $\mu(0)=\sigma/(\sigma-1)$ and $\sigma>\eta\geq 1$.

So *how* firms compete is a parameter, not a commitment. It affects only the magnitude in the paragraph below.

Profit. With $L(S)\equiv 1-1/\mu(S)$ the margin and E market spending,

$$\Pi_g = E\, S_g\, L(S_g) \qquad \text{and, under quantity competition,} \qquad \Pi_g = \frac{E}{\sigma} S_g\big(1+\kappa S_g\big), \quad \kappa \equiv \frac{\sigma}{\eta}-1. \tag{2}$$

Profit rises *faster* than size, and the S^2 term is the entire content of the paper. Its coefficient is $\kappa_{\text{quantity}} = \sigma/\eta - 1 = 4.00$ versus $\kappa_{\text{price}} = (\sigma-\eta)/\sigma = 0.80$ at $\sigma=5, \eta=1$: conduct changes the *magnitude* five-fold and nothing structural.

The two objects that make the model tractable. Define

$$K_g \equiv \sum_{i\in g} c_i^{1-\sigma} \;\; \text{(the group's \textit{strength})}, \qquad A \equiv \sum_j p_j^{1-\sigma} \;\; \text{(how \textit{tough} the market is)}, \qquad \psi(S) \equiv S\,\mu(S)^{\sigma-1},$$

where c_i is factory i's delivered cost. Then market clearing $\sum_g S_g = 1$ plus $p_i = \mu(S_g)c_i$ give

$$\boxed{S_g = \psi^{-1}(K_g/A)} \tag{3}$$

Why this matters. However many factories a group owns, its entire position is the single number K_g — cannibalisation is *already inside it* — and all it needs to know about rivals is the single number A. A market with hundreds of factories is a problem in *one* unknown. That is what makes both theorems possible.

4 The world

Factory a, group headquartered in h, producing in ℓ, selling to n in sector k:

$$c_{a,n} = \frac{1}{\varphi_a} \underbrace{w_\ell^{1-\nu_k}}_{\text{(i) wages}} \underbrace{\big(P^{\text{IO}}_{\ell k}\big)^{\nu_k}}_{\text{(ii) inputs}} \underbrace{\gamma_{h\ell}}_{\text{(iii) away from HQ}} \underbrace{d_{\ell n}}_{\text{(iv) shipping, from } \ell} \underbrace{(1+t_{\ell n k})}_{\text{(v) tariff}} \tag{4}$$

(i) labour hired where the factory sits; (ii) inputs bought where the factory sits; (iii) producing away from headquarters is less efficient; (iv) **shipping counted from the factory, not headquarters**, so one factory serves many countries; (v) tariffs. φ_a is productivity, and it is where being a multinational shows up: head-office services are a non-rival *capability*, so the complexity index α_k raises a stand-alone local firm's penalty and a group's return to capability inside φ_a, without putting any foreign wage into the cost bundle. The variant in which a share α_k of cost is instead head-office work bought at the *parent's* wage is supported and reported alongside; Section 6 explains why it is not the baseline.

Entry. Selling into a market costs a fixed $F_{a,n} = f_k\, w_\ell$ per period (times a distance multiplier), paid in the *same labour as production.* That keeps the model scale-invariant in wages, makes entry costs real resources appearing in labour demand, and makes distributed profits *net* of them.

Income. With θ_{gn} the fraction of group g owned by country n,

$$X_n = \underbrace{w_n L_n}_{\text{wages}} + \underbrace{\textstyle\sum_g \theta_{gn}\big(\Pi_g - \text{entry costs}_g\big)}_{\text{profits owned, net}} + \underbrace{T_n}_{\text{tariffs}}. \tag{5}$$

Profits are earned where factories sell but *received* where owners live; θ separates the two, and varies *firm by firm* rather than being one number per country. That distinction is the paper.

Definition 1 (Equilibrium)**.** Wages, price indices, market sizes, incomes, an active set of factories and quantities such that: (i) every group best-responds in every market; (ii) $\sum_g S_g = 1$ in every market; (iii) no group wants to open or close a factory; (iv) price indices are consistent with the costs they generate; (v) every worker is employed; (vi) every country's budget balances. One of (v) is redundant, so one wage is the numeraire.

5 The mathematical results

Theorem 1 (A market has exactly one equilibrium)**.** *Under Assumption 1, for any set of operating factories belonging to at least two groups, the market equilibrium exists and is unique.*

Proof idea. ψ is a product of two positive increasing functions, hence strictly increasing and invertible. So $T(A) \equiv \sum_g \psi^{-1}(K_g/A)$ is continuous and *strictly decreasing*; it is below 1 for large A and above 1 before A gets small enough for one group to take the whole market. A strictly decreasing continuous curve crosses the line $T = 1$ exactly once. Everything else follows from A by one-way formulas. Separately, each group's profit is strictly concave in its own quantities (revenue concave in the group's output index, cost convex in it), so the first-order conditions identify best responses rather than merely being necessary. □

Entry: the hard part

Entry games normally have *many* equilibria, and the two standard fixes fail here: ranking entrants by cost needs each entrant to be a separate firm (it breaks once a group owns several factories, since switching on its own factory *raises* its share), and imposing an order of moves makes market structure an artifact of the order — fatal when market structure is what we are explaining. The resolution is to separate the firm's effect on *itself* from its rivals' effect on it. Write the entry payoff as $\Omega(K; A) = \Pi(\psi^{-1}(K/A))$, and let $G \equiv \Pi'/\psi'$, $\varepsilon_G \equiv SG'/G$, $\varepsilon_\psi \equiv S\psi'/\psi$. Then, with $a \equiv 2(\sigma - 1)$ and $b \equiv \sigma - 2$, under quantity competition with $\eta = 1$:

$$\textbf{(A) diminishing own returns:} \quad \Omega_{KK} < 0 \iff \varepsilon_G < 0, \qquad \varepsilon_G(S) = S\left[\tfrac{a}{1+aS} - \tfrac{b}{1+bS} - \tfrac{\sigma}{1-S}\right] \tag{6}$$

$$\textbf{(B) strategic substitutes:} \quad \Omega_{KA} < 0 \iff \varepsilon_G + \varepsilon_\psi > 0, \qquad \varepsilon_G + \varepsilon_\psi = \tfrac{aS}{1+aS} - \tfrac{bS}{1+bS} + \tfrac{1-2S}{1-S} \tag{7}$$

Both are proved, not assumed. *(A)* holds at *every* share for $\sigma \geq 2$: the bracket in (6) equals 0 at $S = 0$ (since $a - b - \sigma = 0$) and is strictly decreasing (since $a > b \geq 0$ implies $a^2/(1 + aS)^2 > b^2/(1 + bS)^2$), so it is negative thereafter. *(B)* holds at *every* $S \leq \frac{1}{2}$ and every $\sigma > 1$: in (7) the first two terms are strictly positive because $x \mapsto x/(1+x)$ increases and $a > b$, and the third is non-negative exactly when $S \leq \frac{1}{2}$. At $S = \frac{1}{2}$ its value is exactly $\boxed{1/\sigma}$ — so at $\sigma = 5$ the condition is 20% from binding.

Theorem 2 (The entry outcome is unique)**.** *Under Assumption 1, if in every market no group's share exceeds one half, the entry equilibrium is unique. No order of moves, no selection rule, and groups still compete with themselves.*

Proof. Step 1. Only the efficient (strength, cost) pairs can be optimal, and they form a chain. For two rungs $K_1 < K_2$, the net gain $D(A) = \int_{K_1}^{K_2} \Omega_K \, dK - (\Phi_2 - \Phi_1)$ is strictly decreasing in A by (B). So the larger rung, once abandoned, is never chosen again: the optimum $K_g^*(A)$ is non-increasing in A. *Step 2.* $A(\{K_g\})$ solving $\sum_g \psi^{-1}(K_g/A) = 1$ is strictly *increasing* in every K_g: raising K_j pushes the sum above 1, and the sum falls in A. *Step 3.* Suppose two equilibria with $A < A'$. Step 1 gives $K_g' \leq K_g$ for all g; Step 2 then gives $A' \leq A$, a contradiction. □

Note. A single-factory local firm is the one-rung case, so the same theorem covers the large international groups and the competitive fringe simultaneously.

The result the model exists to produce

Proposition 1 (The correct adjustment is a concentration measure, not an average)**.** *Under quantity competition, with* $\bar{\theta} = \sum_g S_g \theta_g$, $\mathrm{HHI} = \sum_g S_g^2$ *and* $\mathrm{Cov}(\theta, S) = \sum_g \theta_g S_g^2 - \bar{\theta}\,\mathrm{HHI}$,

$$\frac{\Pi_H}{E} = \frac{1}{\sigma}\Big[\bar{\theta} + \kappa\big(\bar{\theta}\,\mathrm{HHI} + \mathrm{Cov}(\theta, S)\big)\Big], \qquad \textit{so} \qquad \boxed{\textit{error of the country-level calculation} = \frac{E\kappa}{\sigma}\,\mathrm{Cov}(\theta, S).} \tag{8}$$

Proof. $L(S) = (1 + \kappa S)/\sigma$ gives $\Pi_H/E = \frac{1}{\sigma}[\sum \theta_g S_g + \kappa \sum \theta_g S_g^2]$; the definition of covariance rewrites the second sum. Setting $\theta_g \equiv 1$ gives total profit $\Pi/E = (1+\kappa\,\mathrm{HHI})/\sigma$, so the country-level calculation is $\bar{\theta}\,\Pi/E$. Subtracting, the $\bar{\theta}$ and $\kappa\bar{\theta}\,\mathrm{HHI}$ terms cancel. □

Reading it. What matters is not what fraction of *firms* a country owns but what fraction of the *big* ones. The bias is the covariance between ownership and size — which cannot be computed from a country-level ownership share, and which the prior literature therefore could not see. This is an identity, not an estimate — and, measured in the matched customs–ownership data (guide, §"The object, measured"), it is far from zero: the country-level calculation understates the United States' ownership-weighted concentration by 10%, overstates Argentina's and Peru's by 24–30%, and the sign varies owner by owner — a pattern no country-level share can express.

Proposition 2 (Ownership irrelevance)**.** *If markups are constant, $\kappa = 0$, both correction terms vanish: given the country-level share $\bar{\theta}$, the* distribution *of ownership across firms is irrelevant to income. Which firms a country owns stops mattering; only how much.*

This is why the standard workhorse model cannot host the question, and why $\kappa > 0$ — i.e. cannibalisation — is not optional.

6 What is proved, and where the proof stops

Layers 1–4 each have exactly one solution, **proved**: the market (Theorem 1); which factories operate (Theorem 2); input prices given wages (a contraction of modulus $\max_k \nu_k < 1$); and spending and incomes (a linear system with $\rho(B) < 1$). Layer 5, wages, is **proved to exist** — excess demand is continuous, scale-invariant, satisfies Walras and diverges as any wage goes to zero — and **proved unique under two conditions that are checked rather than assumed**.

Theorem 3 (Equilibrium wages are unique; Theorem 4 of the guide)**.** *Write $M_{nj} = \partial \ln W_n/\partial \ln w_j$ for the elasticity of country n's wage bill; homogeneity makes the rows of M sum to one, so gross substitutes is exactly $M_{nj} > 0$ for $n \neq j$. Suppose (H1) no group holds more than half of any market, and (H2) for every ordered pair $n \neq m$ the growth of the markets n earns in exceeds n's income-weighted exposure to w_m. Then gross substitutes holds and there is at most one equilibrium wage vector in the region, up to scaling.*

The two lemmas that make it work. *(i) Pass-through.* $\partial \ln P/\partial \ln w$ is non-negative with unit row sums — unconditionally. At every step the response is a weighted average of the responses one level down, so the markup channel, the input–output loop and the head-office linkage are all absorbed without changing the sign. *(ii) Decomposition.* A factory's factor bill responds as market growth, minus $(\sigma - 1)$ times its cost change relative to its own group, minus $[1 - \varepsilon_\mu(S_g)]\chi_g$ times its group's relative to the market. The second coefficient is non-negative exactly when $\varepsilon_\mu \leq 1$, which under quantity competition with $\eta = 1$ is exactly $S_g \leq \frac{1}{2}$ — **the same one half as Theorem 2**. And because a parent's country is paid a fixed share α_k of the group's *whole* bill, the first term averages to zero over a group: the head-office linkage, long suspected of being the obstacle, cancels.

The conditions, at the calibrated five-country economy. (H1): largest parent share 0.344 (quantities) and 0.420 (prices), against 0.50. (H2): satisfied for all 20 ordered country pairs under both conducts, tightest margin 0.024 and 0.025. Gross substitutes itself: 20 of 20 off-diagonals positive, smallest +0.099. Every analytic step is machine-checked against numerical differentiation of the solved model to 2×10^{-7}, the error of the difference quotient.

Where it stops, precisely. (H2) cannot be dropped. In a three-country economy where one country lives entirely off head-office payments for factories located in a second, its wage bill *falls* when the host's wage rises, and gross substitutes fails at the equilibrium — with (H1) satisfied throughout, the largest parent share running from 0.18 to 0.33 against a threshold of 0.50. The statistic that separates that economy from this one is bilateral exposure: 1.00 there, at most 0.38 (quantities) and 0.49 (prices) here. It is not that α_k is too large; gross substitutes survives $\alpha_k = 0.9$ in ordinary geographies, and inside the counterexample family the failure is worst at $\alpha_k = 0.15$ and gone by $\alpha_k = 0.80$. What breaks it is a country whose income is concentrated on a single foreign partner. What uniqueness actually requires is weaker still — index +1 at every equilibrium, which Walras' law makes independent of the numeraire — and in 50 economies from that family, with gross substitutes failing in 34, the index is +1 in all 50 and 25 dispersed starts find one equilibrium in each. Proving the index condition directly is the remaining open problem.

Uniqueness, constructively — the statement to quote. Given wages, everything else in the model is already a unique function of them, so the whole model is one map on wages: the tâtonnement $\Psi_\kappa(w)_n = w_n(\mathrm{LD}_n(w)/L_n)^\kappa$ that the solver iterates. It is homogeneous of degree one, so it acts on the projective space of wage vectors, and its log-Jacobian $(1-\kappa)I + \kappa M$ has unit row sums by that homogeneity. Where that matrix is entrywise positive — gross substitutes off the diagonal, and $\kappa < 1/(1 - \min_n M_{nn})$ on it — Birkhoff's theorem makes Ψ_κ a strict contraction in Hilbert's metric. **One contraction, one fixed point, and the algorithm converges to it from any start in the region where the positivity holds** — that positivity is checked, not proved in general, which is the honest qualifier; but in the $\nu = 0$ version it is now *computer-certified on the continuum* of every wage vector within ±15% (log) of the solution, by outward-rounded interval arithmetic, so there the uniqueness statement carries no between-points gap. At the calibrated five-country economy the contraction modulus is 0.822 and the map still contracts on a box of relative wages of spread 2.0, under both ways of competing; taking random pairs of wage vectors at spread 0.8, the Hilbert distance shrank by at most 0.43 against a certified bound of 0.87. The damping is not a numerical convenience: labour demand is elastic, $M_{nn} \approx -1.49$, so the undamped map is not even monotone, and the model itself says $\kappa < 0.402$ is needed — the solver uses 0.25.

Why the baseline does not need (H2) — Theorem 5 of the guide. α_k does two jobs: it can be a share of a factory's cost paid at its *parent's* wage, and it indexes complexity in the productivity draws. Only the first creates the cross-country factor linkage, and it is the whole content of the counterexample. The baseline makes head-office services a non-rival *capability* — keeping the second job, dropping the first — and that buys Theorem 5: under (H1) alone the reallocation block of $\partial \ln W_n/\partial \ln w_m$ is non-negative, so gross substitutes reduces to a pure demand condition. Relative to the cost-share variant, the calibrated floor moves from −1.79 to +0.014 (at $\nu = 0$) and the gross-substitutes margin widens from +0.100 to +0.360; on adversarial economies satisfying (H1) the cost-share variant violates gross substitutes at 19 of 250 points and the baseline at 0 of 340. **And it fits better**: Fact 2's gradient goes from +0.33 to +0.42 against a target of +0.43, Fact 1 is unchanged, and the fitted multinational productivity edge falls to zero.

The old evidence still stands, and it checks what the theorem cannot, because the theorem holds the operating set fixed: 10 dispersed starts reach wages agreeing to 1.5×10^{-11}, labour-market error 6.5×10^{-12}, *and the same set of operating firms.* Entry is also in whole numbers, so an exact integer solution need not exist; the residue is 1 slot of 1,050, falling with scale — a near-miss on *existence*, not multiplicity.

Accounting. All identities (sales = wages + inputs + tariffs + profit; wage bill = production + entry-cost labour; Walras; world income = world spending; country budgets) hold to 2.4×10^{-15} at wages far from equilibrium and under both conducts. Largest group share at the solution: 0.344 (quantities), 0.420 (prices) — both inside the $\frac{1}{2}$ bound of Theorem 2.

7 Model versus data

#	Fact	Model	Data	Status
1	MNE export share; foreign dominant	0.69 (0.54 / 0.15)	0.46–0.74	level fitted; **split produced**
2	foreign firms in complex goods	+0.42	+0.43	fitted; *local half unresolved*
3	parents from few countries	0.273	0.130	**produced**, overshoots
4	grouping by owner raises concentration	×1.94	×1.12	**produced** — a prediction
5	MNEs present ⇒ local exports	−100%	positive	**fails**, worse with entry
6	distance matters less for MNEs	−0.52/+0.70	−0.16/+0.05	**produced**, too large

Three parameters are fitted (entry cost to the *number of exporters*; head-office disadvantage to Fact 2's gradient; pool scaling to Fact 1's split), so Facts 3, 4 and 6 are genuine predictions. There used to be a fourth, a multinational productivity edge fitted to Fact 1's level; once head-office services are a capability rather than a factor bought at the parent's wage it is no longer needed and is set to zero. **Fact 5 is the one outright failure** and belongs in the abstract. The suspicion that the estimate was contaminated — its controls omitted a destination-by-product-by-year term — has been tested directly: with that term added, the coefficient halves (0.166 → 0.087, both margins, PPML agreeing) but survives, precisely estimated with the same sign. The fact is real at half its published size. And it cannot be fitted away: a productivity spillover moves the model's response only from −97% to −60% at indefensible sizes, and a *fixed-cost* spillover that fully repairs the entry margin — every local plant surviving the influx — still leaves −80%, because with market spending fixed, entry is zero-sum in value. The mechanism must instead expand the origin's *penetration* of the destination market against a rest-of-world supply — the λ margin of the measurement operator. That outside supplier is now built into the general equilibrium (uniqueness untouched), and it works: it is the first channel to move the fact at first order, the *extensive* margin flips to the data's sign, and at the data's $\lambda \approx 0.09$ the response moves from ≈ -2.9 to ≈ -0.2 per log-unit of international presence, against the measured +0.087. What remains is a joint recalibration with the margin on — the model's next task.